

Probing the dark energy in the functional protein universe

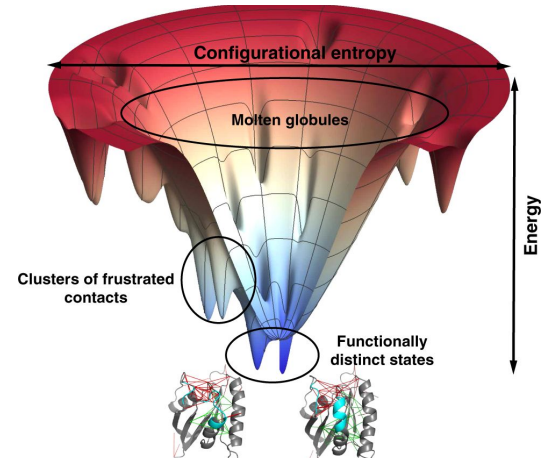
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Background

- Protein folding satisfies the **principle of minimal frustration**: a funnel-like energy landscape.
- Sequence covariation suggests evolutionary and folding landscapes largely correspond, though not perfectly.
- Function (catalysis, binding, specificity) may be at a trade-off locally (e.g. at the binding site) with folding stability.
- Mega-scale stability scans (Deep Mutational Scans) and protein language models (ESM-2) now allow both landscapes (folding vs. evolutionary) to be scored per variant.



Depiction of principle of minimal frustration
(Image not from this paper)

Background

- **Protein Universe** is the collection of all possible protein folds, partially discovered by the biosphere.
- Each fold have multiple sequences specifying many local chemical interactions and functions.
- Following the cosmology analogy:
 - Dark regions of protein universe:** the folds and functions that are not well characterized and understood by existing methods.
 - Dark energy:** Discrepancy between folding landscape and evolutionary landscape.
- When the dark energy is quantified we have measured “the strength of the functional constraints”.

Folding vs evolutionary landscapes

- Sequences (σ) in a family follow a Boltzmann-like distribution. Ψ_{evo} here is “Evolutionary Potential” and can be inferred by DCA, VAEs, transformers or pLMs.

$$p(\sigma) = \frac{e^{-\Psi^{evo}(\sigma)}}{Z}.$$

- From the fold perspective, a similar image: $p(\sigma) = \frac{e^{-\frac{E^{fold}(\sigma)}{k_B T_{sel}}}}{Z},$

- If folding were the only constraint, Ψ_{evo} would be a rescaled folding energy: $\Psi^{evo}(\sigma) = \frac{E^{fold}(\sigma)}{k_B T_{sel}}.$
- T_{sel} , the selection temperature, quantifies how strongly folding stability acts as an evolutionary force, in other words: how stable the evolution selects the structures.
 - This is not a real temperature in physical sense. It is similar to the temperature in language models where “less likely” answers are more often supplied.
- The two landscapes agree globally but disagree locally.

Defining dark energy

The difference between free evolutionary energy (rescaled by the folding selection temperature) and the folding energy: the functional constraints beyond folding.

$$E^{dark}(\sigma) = k_B T_{sel}^{fold} \Psi^{evo}(\sigma) - E^{fold}(\sigma)$$

$$p(\sigma) = \frac{e^{-\left[E^{fold}(\sigma)/k_B T_{sel}^{fold} + E^{func}(\sigma)/k_B T_{sel}^{func} \right]}}{Z}.$$

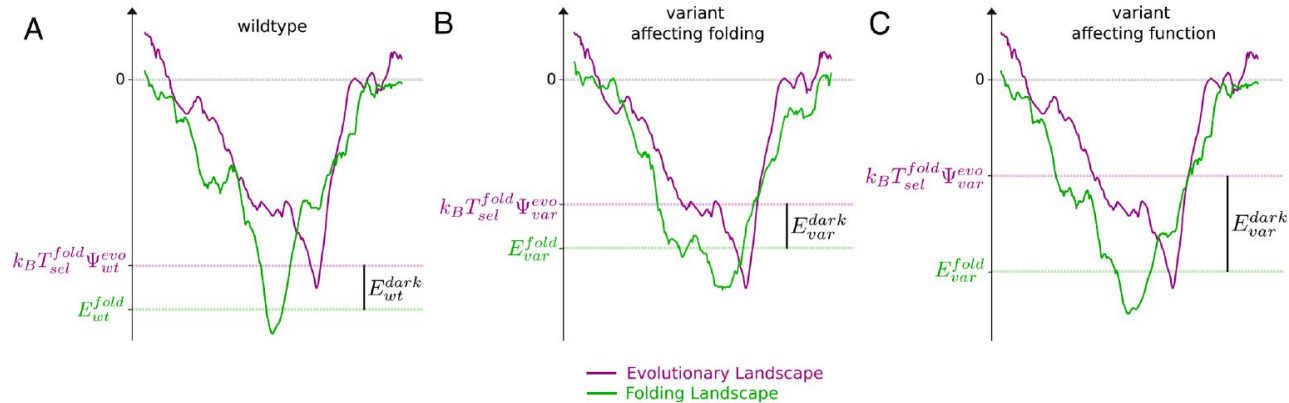


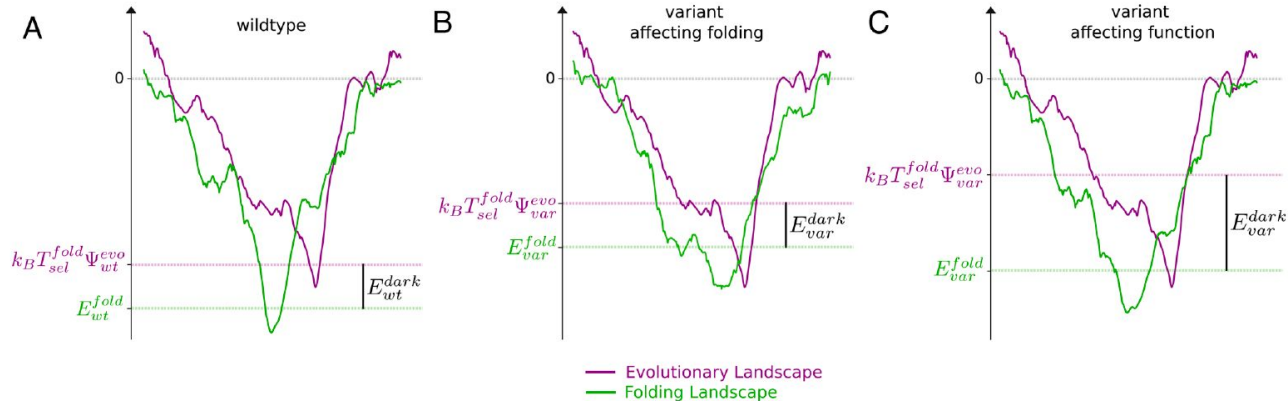
Fig. 1 · (A) wild type, (B) variant affecting folding: dark energy preserved, (C) variant affecting function: dark energy increases

$$T_{sel}^{func} = T_{sel}^{fold} \frac{E^{func}}{E^{dark}}.$$

Dark energy change of single-site variants

$$\Delta E^{dark} = k_B T_{sel}^{fold} \Delta \Psi^{evo} - \Delta E^{fold}$$

- Notice the delta, most variants that have deleterious effects decrease the folding stability (panel B).
- If the variation effects more than folding (some function too): delta E_{dark} is positive (panel C)
- (Consider the formula and how energies affect the values.)



Dark energy change of single-site variants: Experimentally

$$\Delta E^{dark} = k_B T_{sel}^{fold} \Delta \Psi^{evo} - \Delta E^{fold}$$

- **$\Delta \Psi^{evo}$** : ESM-2 (650M) masked-marginal log-likelihood ratio, mutant vs wild type; higher = less likely.
- **ΔE^{fold}** : experimental $\Delta \Delta G$ (Rocklin mega-scale dataset) or computed with the coarse-grained AWSEM [Associative memory, Water mediated, Structure and Energy Model] force field.
- Notice the delta everywhere!
- $k_B T_{sel}^{fold}$ is just a scaling factor for getting both energies to same unit, the ratio of standard deviations:

$$k_B T_{sel}^{fold} = \text{sd}(\Delta \Delta G) / \text{sd}(\Delta \Psi^{evo})$$

Datasets

Question	Data	Δ Efold source
Folding stability survey	64 small domains, Mega-scale folding stability (via ProteinGym)	Experimental $\Delta\Delta G$; AWSEM check
Enzymes	Catalytic Site Atlas, 732 PDBs	AWSEM
Binding	SKEMPI: single-partner complexes with ≥ 5 single-site mutants (32 cases)	AWSEM, plus $\Delta\Delta G$ of binding

$\Delta\Psi_{\text{evo}}$ from ESM-2 throughout.

Case study: FKBP N-terminal domain (2KFV)

- Most variants lie along the *TseI* line; outliers (red) are deleterious without matching destabilization.
- High ΔE_{dark} concentrates at specific positions and substitutions. (Notice the red strips)
- These sites cluster spatially in the region with chemical shift changes on DNA binding.
- Dataset average $T_{\text{seI}} = 156 \pm 9$ K, below T_f .

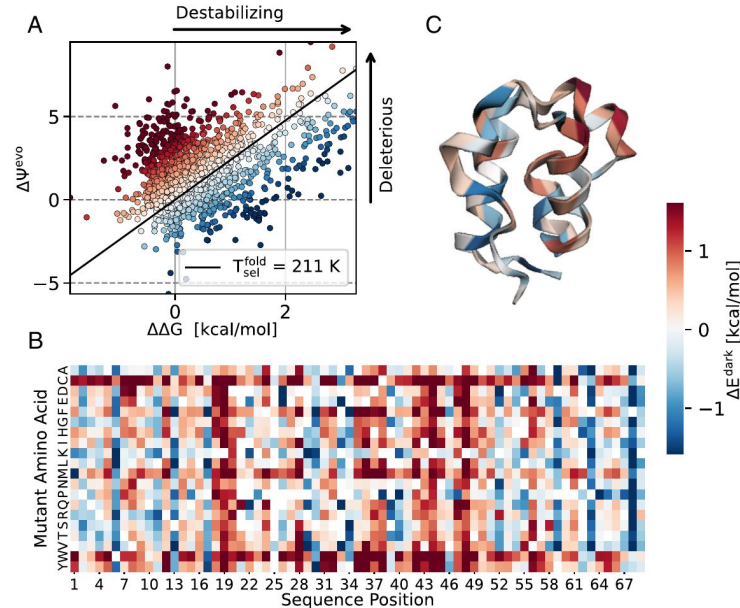


Fig. 2 · (A) $\Delta\Psi_{\text{evo}}$ vs $\Delta\Delta G$, (B) mutational heatmap, (C) site-averaged ΔE_{dark} on structure

Survey of 64 domains

- High dark energy sites appear in helices, sheets and loops.
- Two-component Gaussian mixture; threshold 1.03 kcal/mol.
 - **74.7%** of variants mainly affect stability, **25.3%** mainly function.

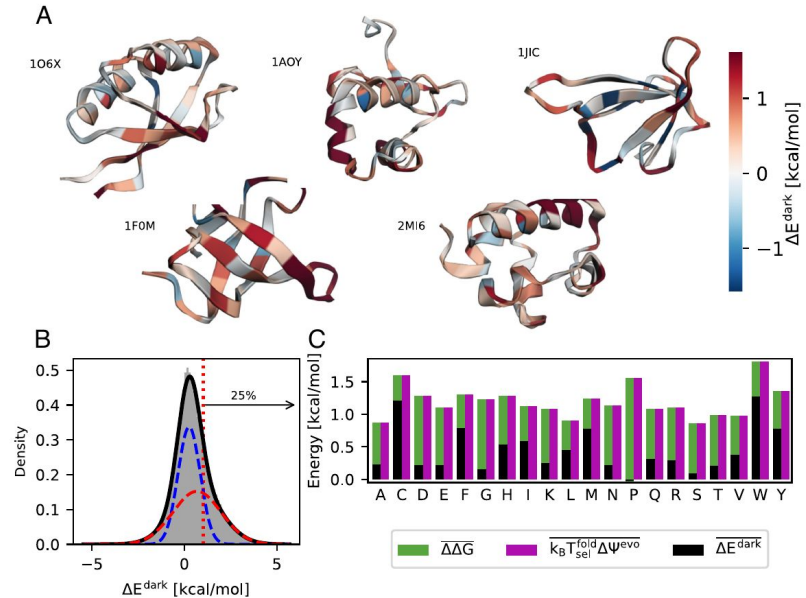


Fig. 3 · (A) structures colored by site-averaged ΔE^{dark} , (B) weighted distribution and GMM, (C) averages per mutant amino acid

Survey of 64 domains

- Mutations to C and W carry the highest ΔE_{dark} ; P is near zero, its effect explained by destabilization.
 - Proline is known to disrupt secondary structures.
 - C and W are relatively less abundant and dark energy observed to be inversely correlated with relative abundance
- Mutations to Alanine (A) is observed to result in “site average” delta dark energy, good target for low-cost mutational scanning for this.

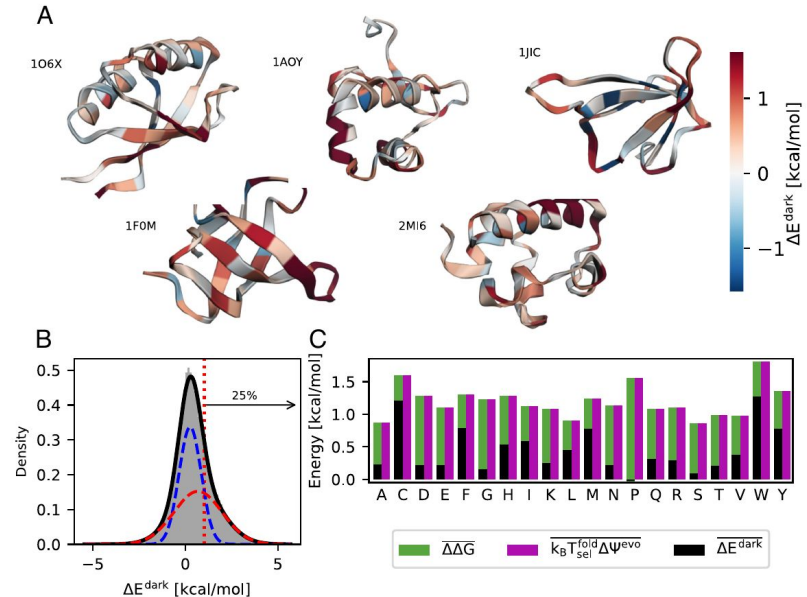


Fig. 3 · (A) structures colored by site-averaged ΔE_{dark} , (B) weighted distribution and GMM, (C) averages per mutant amino acid

Replacing experiments with AWSEM

- ΔE_{fold} recomputed with AWSEM (burial + contact terms, native structure fixed), rescaled to kcal/mol.
- ΔE_{dark} correlates with the experimental version: average Pearson $r = 0.63$.
- Threshold 1.36 kcal/mol; 36.4% of variants classed as functional, possibly reflecting force-field limits. (it was ~25% experimentally)
- Site-averaged patterns are preserved, so the method extends to proteins without stability scans.

SI Appendix, Fig. S4

Dark energy in enzymes

- TEM-1 β -lactamase: catalytic residues (S70, K73, S130, E166, K234) exceed 3 kcal/mol (very high).
- A halo of dark energy around the active site, decaying with distance.
- Functional fraction by shell: active 87.6%, inner 1 66.8%, inner 2 60.7%, outer 38.9%.
- Restabilizing mutants M182T and L201P show negative ΔE_{dark} . (Gain-of-function)
- “The same patterns were found across all the enzyme classes of the studied dataset (SI Appendix, Fig. S5)”
- High dark energy far away from active sites may signal allostery

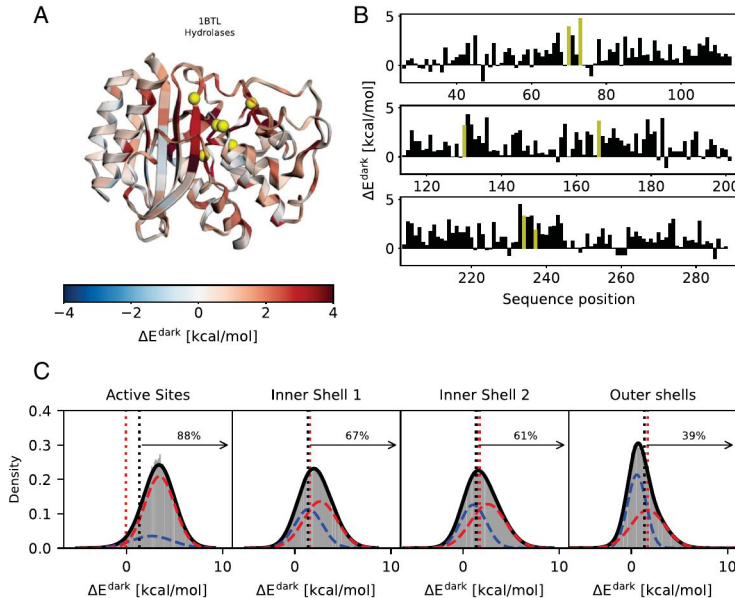


Fig. 4 • (A, B) TEM-1 (1BTL), active sites in yellow, (C) Catalytic Site Atlas distributions by shell

Binding selection temperature

$$T_{sel}^{func} = T_{sel}^{fold} \Delta E^{func} / \Delta E^{dark}$$

- Barstar-Barnase: ΔE^{dark} localizes to the interface; $\Delta\Delta G_{binding}$ vs ΔE^{dark} $r = 0.87$.
- Basically -> We expect functional effects to be proportional to dark energy
- $T_{sel}^{fold} = 141$ K, $T_{sel}^{bind} = 186 \pm 31$ K.
- Across SKEMPI cases with $r > 0.5$, T_{sel}^{bind} ranges 60–186 K.

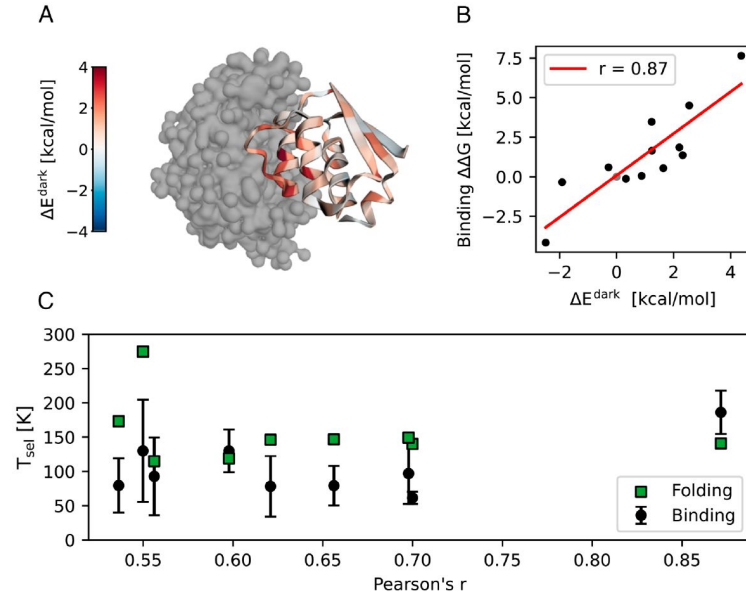


Fig. 5 · (A) Barstar on Barnase (1BRS), (B) binding $\Delta\Delta G$ vs ΔE^{dark} , (C) T_{sel} for folding and binding vs Pearson's r

Takeaways

- Dark energy is a variant-level score in physical energy units, comparable across proteins.
- About 25–35% of positions carry significant dark energy, roughly independent of protein size.
- It localizes at functional sites: binding regions, catalytic centers and their surrounding shells.
- Selection for binding is about as strong as selection for folding, but binding alone explains dark energy in only about a third of the systems.

Discussion

- Dark energy is a variant-level score in physical energy units, comparable across proteins.
- About 25–35% of positions carry significant dark energy, roughly independent of protein size.
- It localizes at functional sites: binding regions, catalytic centers and their surrounding shells.
- Selection for binding is about as strong as selection for folding, but binding alone explains dark energy in only about a third of the systems.
- Caveats noted by the authors: one function beyond folding is an oversimplification; misfolded-state energy approximation; AWSEM force-field inaccuracies; sparse binding data.
- How much does the result depend on ESM-2 as the proxy for Ψ_{evo} ?
- Can dark energy flag function where the function is unknown?

An information-theoretic argument for the restriction of the current biological alphabets to 4 nucleotides and 20 amino acids

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The question

- Life is built on foldable biopolymers coded with just 4 nucleotides or 20 amino acids, out of many chemically plausible alternatives.
- Evolution needs a fast search of two spaces: conformations (folding) and sequences (evolution).
- Energy landscape theory links sequence and conformation free energies to foldability.
- Molecular information theory links conformational entropy to sequence entropy.
- Combining the two constrains the alphabet size of an evolving biopolymer.

Information gained by folding = information in the sequence

$$R_{\text{Levinthal}} = \log_2 \left(\frac{N_{\text{unfolded}}}{N_{\text{native}}} \right) = R_{\text{sequence}} = \log_2 \left(\frac{\text{Alphabet size}}{N_{\text{evolved}}} \right)$$

- **R Levinthal:** entropy lost per site when finding the native state among unfolded alternatives. Empirically 2.4 ± 0.3 bits/(site·operation) for proteins. Can be defined per site, amino acid, per protein.
- **R sequence:** entropy lost per site going from random sequences to the evolved ensemble compatible with the structure. Empirically 2.1 ± 0.3 bits/(site·operation). (this is -> PLM loss function cross entropy loss)
- N = effective number of conformations (unfolded, native) or monomers (evolved) per site.

$$\frac{\text{Alphabet size}}{N_{\text{evolved}}} = \frac{N_{\text{unfolded}}}{N_{\text{native}}}$$

The energy landscape condition

- Random energy model: foldable sequences exist for a target structure only if the sequence-space entropy is positive.

The logarithm of that count, per site, is the "entropy in sequence space for a given target structure":

$$s \approx \ln N_{\text{evolved}} - \ln N_{\text{unfolded}}$$

- **If $s > 0$** , the count grows exponentially with chain length. Many sequences encode the fold, and evolution can find them and drift among them.
- That holds when evolved sequences tolerate more monomers per site than the unfolded chain has conformations per site:

$$N_{\text{evolved}} > N_{\text{unfolded}}$$

Combining the two

The equality in Eq. 1 ties the alphabet size to the other three quantities:

$$\frac{\text{Alphabet size}}{N_{\text{evolved}}} = \frac{N_{\text{unfolded}}}{N_{\text{native}}}$$

At the limit of Eq. 3, $N_{\text{evolved}} = N_{\text{unfolded}}$:

$$(N_{\text{unfolded}})^2 = \text{Alphabet size} \cdot N_{\text{native}} = (N_{\text{evolved}})^2$$

Extant folded biopolymers have a single reference structure, $N_{\text{native}} \approx 1$:

$$(N_{\text{unfolded}})^2 = \text{Alphabet size} = (N_{\text{evolved}})^2$$

Predictions

- N unfolded is set by the polymer-solvent physical chemistry; given it, Eq. 5 fixes both the alphabet size and N evolved.
- Three testable relations: N unfolded = N evolved = $\sqrt{\text{alphabet size}}$.
- RNA (4 letters): N unfolded = N evolved = 2. Proteins (20 letters): ≈ 4.47 .
- Information an evolved polymer can encode:

$$R_{\text{sequence}} = \log_2 \left(\sqrt{\text{Alphabet size}} \right)$$

1 bit/site for RNA, about 2.2 bits/site for proteins.

Empirical estimates

	Predicted	N unfolded	N evolved
RNA	2	2.54 ± 0.25 (23S backbone entropy 2.13; ribozymes/riboswitches 3.00; base-pair alphabet 2.48)	2.35 ± 0.024 (over 1200 Rfam alignments)
Proteins	4.47	5.38 ± 0.46 (six estimates: backbone entropy, secondary structure alphabet, burial, contacts)	4.82 ± 0.32 (seven sequence entropy estimates)

RNA backbone entropy converted with $N_{\text{unfolded}} = e^{\Delta S/R}$

Theory vs data

- N unfolded (blue) is close to $\sqrt{(\text{alphabet size})}$ for both RNA and proteins.
- N evolved (red) is close to $\sqrt{(\text{alphabet size})}$ for both.
- N evolved $\approx$ N unfolded, so the sequence space search is also efficient.
- All three predictions of Eq. 5 hold.

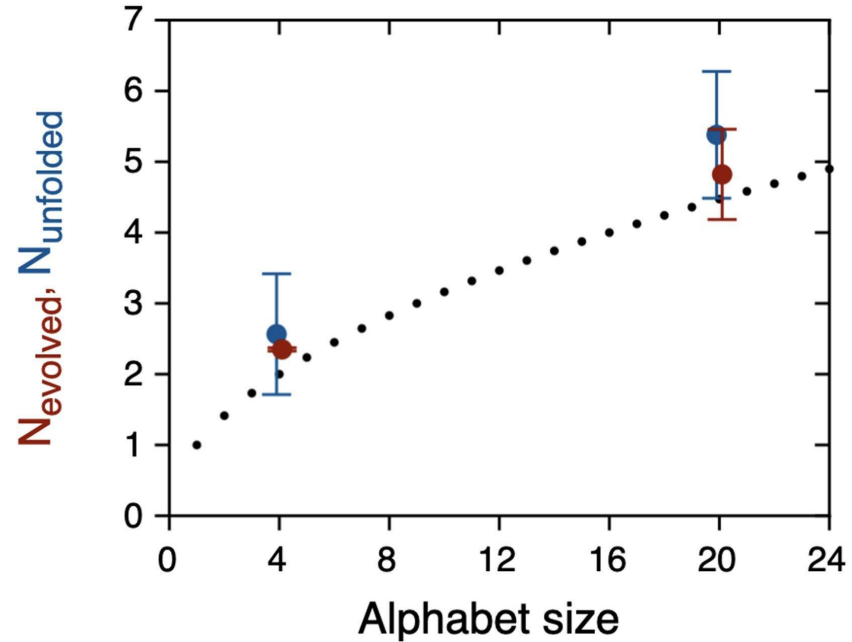


Fig. 1 · black: Eq. 5 prediction; red: N evolved; blue: N unfolded; error bars: 95% CI of the mean

Evolving a foldable biopolymer

- Start near random: $N_{\text{evolved}} \approx \text{alphabet size}$, $R_{\text{sequence}} \approx 0$.
- Selection for folding lowers N_{evolved} and raises R_{sequence} .
- At $N_{\text{evolved}} = N_{\text{unfolded}}$, R_{sequence} matches the required $R_{\text{Levinthal}}$.
- Going further would create excess information that dissipates (entropy) and make the foldable-sequence search unfeasible (Eq. 3).
- So natural sequences should sit near the limit $N_{\text{evolved}} \approx N_{\text{unfolded}}$.
- Therefore $\rightarrow$ biopolymer chemical-physics (N_{unfolded}) strongly constraints the alphabet size.

Early life: trading alphabet size for disorder

Eq. 4 makes $N_{\text{native}} \cdot \text{alphabet size}$ a constant for a given chemistry. With N_{native} between 1 and N_{unfolded} :

$$N_{\text{native}} \in [1, N_{\text{evolved}} = N_{\text{unfolded}}]$$

$$\text{Alphabet size} \in \left[N_{\text{evolved}} = N_{\text{unfolded}}, (N_{\text{unfolded}})^2 = (N_{\text{evolved}})^2 \right]$$

A smaller alphabet can still code evolvable proteins, but with more conformations per site in the native state.

A scenario for the genetic code (not experimental)

- Early alphabets of 2–3 amino acids: below the ≈ 4.47 minimum for foldable proteins.
- Jump to 7 (G, A, V, D, E, P, S) near $t \approx 7.5$: N native ≈ 3 , consistent with intrinsically disordered proteins; 5 of 7 are disorder-promoting.
- $t \approx 7.5$ –17: steady growth to 20, mostly order-promoting residues; N native falls toward 1.

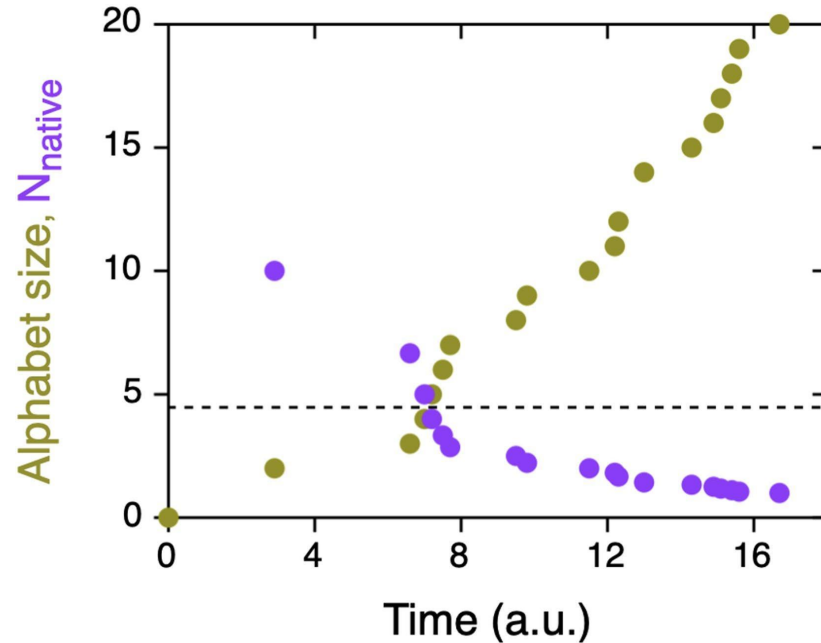


Fig. 2 · green: alphabet size (Trifonov 2000); purple: N native from Eq. 4; dashed: minimum alphabet ≈ 4.47

Discussion

- At the limit, one effective digital letter maps to one effective analog conformation: $N_{\text{evolved}} = N_{\text{unfolded}}$.
 - Thus, simplest codable & foldable polymer can have $N=2$ (1 bit).
 - The case of RNA: $N_{\text{unfolded}} \approx 2$, so 4 letters is the minimum.
- Proteins: $N_{\text{unfolded}} = 5.38 \pm 0.46$, allowing 20–39 sufficiently different amino acids; 20 sits at the lower end.
 - The reasons suggested: synthesis machinery, post-translational modification, functional substates.
 - Here dark energy paper is mentioned. Some positions may encode more information than simply folding.
- Early code evolution likely went from low-complexity peptides to disordered but evolvable proteins, then to well-folded ones.
- With $N_{\text{native}} = 1$ reachable at 20 letters, the genetic code may be largely complete.
- Note that the early-life scenario rests on a semi-quantitative, ad hoc time scale and is speculative.

Thanks for listening.



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saying Howdy?